

SYDNEY HORSLEY PARK, Australia

WMO: 947600

Lat: 33.8511S

Lon: 150.8567E

Elev: 100

StdP: 100.13

Time Zone: 10.00 (AUS)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 3.5	(c) 4.8	(d) -5.1	(e) 2.5	(f) 19.7	(g) -3.0	(h) 3.0	(i) 17.3	(j) 9.3	(k) 16.8	(l) 8.0	(m) 15.3	(n) 0.8	(o) 310	(p) 0.388

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.8	35.5	20.1	32.9	20.2	30.6	19.7	23.1	29.8	22.3	28.2	21.6	26.9	4.1	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.5	16.3	24.7	20.8	15.6	24.0	20.1	14.9	23.4	68.9	29.8	65.9	28.3	63.5	26.9	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.1	(b) 7.4	(c) 6.4	(e) 0.6	(f) 40.8	(g) 1.0	(h) 2.6	(i) -0.1	(j) 42.7	(k) -0.7	(l) 44.2	(m) -1.3	(n) 45.7	(o) -2.0	(p) 47.6		
(a) 9.1	(b) 7.4	(c) 6.4	(e) 0.0	(f) 24.8	(g) 1.0	(h) 0.9	(i) -0.7	(j) 25.4	(k) -1.4	(l) 25.9	(m) -1.9	(n) 26.4	(o) -2.7	(p) 27.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	23.5	22.7	21.1	18.2	14.7	12.4	11.6	12.7	15.5	17.9	20.0	21.8		
	DBStd	4.84	3.04	2.78	2.43	2.37	2.13	1.91	2.03	2.22	2.84	3.11	3.24	3.09		
	HDD10.0	11	0	0	0	0	0	3	6	2	0	0	0	0		
	HDD18.3	878	1	1	4	29	113	177	210	177	96	47	17	6		
	CDD10.0	2803	419	354	344	247	147	76	55	84	164	245	301	366		
	CDD18.3	626	162	122	89	27	1	0	0	1	9	34	68	113		
	CDH23.3	4717	1263	803	472	152	11	0	1	7	137	356	609	906		
	CDH26.7	1822	566	302	130	36	0	0	0	0	36	123	254	377		
(14) Wind	WSAvg	2.6	3.0	2.8	2.5	2.3	2.2	2.4	2.3	2.5	2.6	2.8	3.0	2.9		
(15) Precipitation	PrecAvg	900	106	104	104	76	70	66	40	54	46	70	90	75		
	PrecMax	1460	305	456	346	364	217	304	144	458	170	196	391	267		
	PrecMin	521	5	7	6	2	3	1	0	0	0	0	11	2		
	PrecStd	239	76	91	77	77	63	71	37	83	38	54	70	58		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.7	37.4	32.9	30.6	24.8	20.7	22.3	24.1	30.3	33.3	37.0	37.5		
		MCWB	21.6	20.6	19.6	17.3	14.9	13.5	13.0	12.7	15.6	16.0	17.1	20.2		
	2%	DB	35.3	33.1	29.8	26.9	22.8	18.8	19.4	21.7	27.1	29.8	32.9	33.8		
		MCWB	21.5	21.8	20.6	17.3	14.7	12.8	11.3	12.2	14.1	15.8	18.4	20.0		
	5%	DB	32.3	30.0	27.8	24.7	21.2	17.6	17.7	19.9	24.4	27.4	29.2	30.7		
		MCWB	21.2	21.1	20.0	17.3	14.3	12.4	11.0	11.8	13.7	15.9	18.3	19.3		
	10%	DB	29.6	27.9	25.8	23.0	19.7	16.6	16.3	18.2	21.9	24.6	26.6	28.2		
		MCWB	20.7	20.3	19.3	16.9	13.6	11.9	10.6	11.0	13.1	15.5	17.9	19.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.5	24.2	23.1	20.5	18.0	16.0	15.5	15.7	17.5	20.0	22.1	23.1		
		MCDB	32.8	31.6	27.7	25.3	20.5	18.4	18.8	20.2	23.5	25.9	30.1	30.6		
	2%	WB	23.2	22.9	21.8	19.3	16.7	14.6	13.6	14.2	16.3	18.7	20.8	22.1		
		MCDB	30.3	29.0	26.1	23.3	19.4	16.4	16.2	17.9	22.1	24.2	27.2	28.9		
	5%	WB	22.4	22.0	21.0	18.6	15.6	13.8	12.5	13.1	15.3	17.8	19.8	21.1		
		MCDB	28.6	27.2	25.1	22.2	19.1	15.8	15.3	17.7	20.9	23.1	25.7	27.0		
	10%	WB	21.7	21.3	20.2	17.8	14.7	13.1	11.5	12.1	14.4	16.9	19.0	20.3		
		MCDB	27.1	25.9	24.2	21.3	18.1	15.4	14.7	16.2	19.7	21.5	24.1	25.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	9.4	9.3	9.7	10.5	8.9	10.4	11.4	12.1	11.6	11.0	11.0		
		MCDBR	16.7	13.9	12.5	12.6	12.7	10.6	12.4	14.1	16.6	16.8	16.9	16.1		
	5% WB	MCWBR	5.0	4.4	4.7	5.0	5.8	5.8	6.3	6.5	6.4	6.0	5.5	5.2		
		MCWBR	12.9	11.2	10.7	10.2	10.1	7.7	9.2	11.2	13.0	12.7	12.9	12.9		
(40) Clear-Sky Solar Irradiance	taub	0.391	0.389	0.359	0.331	0.315	0.306	0.296	0.303	0.335	0.356	0.381	0.391	0.391		
		taud	2.446	2.472	2.539	2.592	2.586	2.610	2.620	2.591	2.499	2.470	2.442	2.424		
	Ebn at Noon	946	927	921	894	856	840	871	911	929	949	950	950	950		
		Edn at Noon	120	113	99	83	74	68	70	82	101	112	120	124		
(44) All-Sky Solar Radiation	RadAvg	6.43	5.50	4.70	3.82	3.03	2.39	2.80	3.70	4.93	5.70	6.04	6.60	6.60		
	RadStd	0.49	0.51	0.42	0.33	0.21	0.21	0.21	0.22	0.29	0.53	0.56	0.59	0.59		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	-2	-85	+184	+98

Nomenclature: See separate page